



IT-SIMPLERA INSTITUTE

Cyber Security Analyst Internship Program

WEEK 02 ASSIGNMENT

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Computer Networking Lab Report

Networking Fundamentals — Commands, Devices & Models

Table of Contents

Introduction 3

Q1. Understanding Your Network Information 3

Q2. Basic Networking Commands 4

Q3. Router vs Switch..... 7

Q4. IPv4 vs IPv6 7

Q5. The OSI Model 8

Network Diagram 10

Conclusion..... 11

Introduction

This report documents a hands-on networking lab covering how to identify a computer's local network identity, use core command-line diagnostic tools, and understand the devices, addressing schemes, and layered model that make modern networks work.

Each section below walks through a specific task — from reading ipconfig output to comparing IPv4 and IPv6 — with screenshots, explanations, and supporting diagrams.

Q1. Understanding Your Network Information

Using the ipconfig /all command, the computer's IPv4 Address, MAC Address, and Default Gateway were identified. The purpose of each is explained below.

IPv4 Address: 192.168.118.1

- Purpose: This is your computer's unique local identity card on your home or office network. It allows other local devices to find and talk to your computer.

MAC Address (Physical Address): 00-50-56-C0-00-08

- Purpose: This is a permanent, physical hardware ID burned into your computer's network card by the manufacturer. It never changes, no matter what network you connect to.

Default Gateway: (Example: 192.168.1.1)

- Purpose: This is the IP address of your router. It acts as the “exit door” your computer uses to send data out to the internet.

Network Information

Q1. Using the **ipconfig /all** command, identify your computer's IPv4 Address, MAC Address, and Default Gateway. Explain the purpose of each and attach Screenshots of your results.

IPv4 Address: (192.168.118.1)

- Purpose: This is your computer's unique local identity card on your home or office network. It allows other local devices to find and talk to your computer.
- **IPv4 Address. : 192.168.118.1(Preferred)**

MAC Address (Physical Address): (00-50-56-C0-00-08)

- Purpose: This is a permanent, physical hardware ID burned into your computer's network card by the manufacturer. It never changes, no matter what network you connect to.
- **Physical Address. : 00-50-56-C0-00-08**

Default Gateway: (Example: 192.168.1.1)

- Purpose: This is the IP address of your router. It acts as the "exit door" your computer uses to send data out to the internet.
- **Default Gateway :**

Figure 1 — ipconfig /all summary: IPv4 Address, Physical Address, and Default Gateway

Q2. Basic Networking Commands

The following networking commands were executed, with screenshots of the output and a short explanation of each.

`ipconfig /all`

```
C:\Users\E Z Z O>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : 
    Primary Dns Suffix . . . . . : 
    Node Type . . . . . : 
    IP Routing Enabled. . . . . : 
    WINS Proxy Enabled. . . . . : 

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . : 
    Description . . . . . : 
    Physical Address. . . . . : 
    DHCP Enabled. . . . . : 
    Autoconfiguration Enabled . . . . : 
    Link-local IPv6 Address . . . . . : 
    IPv4 Address. . . . . : 
    Subnet Mask . . . . . : 
    Lease Obtained. . . . . : 
    Lease Expires . . . . . : 
    Default Gateway . . . . . : 

    DHCP Server . . . . . : 
    DHCPv6 IAID . . . . . : 
    DHCPv6 Client DUID. . . . . : 
    DNS Servers . . . . . : 
    NetBIOS over Tcpip. . . . . :
```

Figure 2 — Full `ipconfig /all` output for VMware virtual network adapters (VMnet1 & VMnet8)

Explanation: Shows all your computer's current network settings, including your IP address, MAC address, and DNS servers.

ping google.com

```
C:\Users\E Z Z O>ping google.com

Pinging google.com [142.250.202.174] with 32 bytes of data:
Reply from 142.250.202.174: bytes=32 time=41ms TTL=116
Reply from 142.250.202.174: bytes=32 time=40ms TTL=116
Request timed out.
Reply from 142.250.202.174: bytes=32 time=40ms TTL=116

Ping statistics for 142.250.202.174:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 40ms, Maximum = 41ms, Average = 40ms
```

Explanation: Tests if your computer can successfully connect to Google's server and measures how fast the connection is in milliseconds.

tracert google.com

```
C:\Users\E Z Z O>tracert google.com

Tracing route to google.com [142.250.202.174]
over a maximum of 30 hops:

  0  142.250.202.174  <1 ms  <1 ms  <1 ms
  1  <1 ms  <1 ms  <1 ms
  2  3 ms  3 ms  3 ms
  3  27 ms  27 ms  26 ms
  4  24 ms  24 ms  24 ms
  5  39 ms  42 ms  39 ms
  6  *  *  *
  7  40 ms  39 ms  39 ms
  8  39 ms  39 ms  39 ms
  9  43 ms  39 ms  54 ms
 10  43 ms  44 ms  42 ms
 11  46 ms  *  41 ms
 12  43 ms  43 ms  *
 13  42 ms  40 ms  41 ms

Trace complete.
```

Explanation: Maps out the exact path (the “hops” or routers) your data travels through to get from your computer to Google.

nslookup google.com

```
C:\Users\E Z Z 0>nslookup google.com
DNS request timed out.
    timeout was 2 seconds.
Server:    UnKnown
Address:   8.8.8.8

Non-authoritative answer:
Name:      google.com
Addresses: 192.168.100.1
```

Explanation: Acts like a phonebook look-up and it converts the human-readable website name (google.com) into its actual numerical IP address.

arp -a

```
C:\Users\E Z Z 0>arp -a

Interface: 192.168.30.
Internet Address
192.168.30.255
224.0.0.22
224.0.0.251
224.0.0.252
239.255.255.250

Interface: 192.168.100
Internet Address
192.168.100.1
192.168.100.255
224.0.0.22
224.0.0.251
224.0.0.252
239.255.255.250
255.255.255.255

Interface: 192.168.118
Internet Address
192.168.118.255
224.0.0.22
224.0.0.251
224.0.0.252
239.255.255.250
```

Explanation: Shows a list of all the local physical hardware addresses (MAC addresses) mapped to IP addresses on your immediate local network.

Q3. Router vs Switch

A comparison of a Router and a Switch across six key points: their functions, OSI layer, addressing, and common use cases.

Comparison Feature	Switch	Router
Primary Function	Connects multiple devices together within the same local network.	Connects multiple different networks together (e.g., your home network to the internet).
OSI Layer	Operates at Layer 2 (Data Link Layer).	Operates at Layer 3 (Network Layer).
Addressing Used	Uses MAC Addresses to move data.	Uses IP Addresses to move data.
Data Format	Deals with data in units called Frames.	Deals with data in units called Packets.
Intelligence	Only knows about devices directly plugged into it.	Smart enough to find the quickest path across the internet to route data.
Common Use Case	Used to connect computers, printers, and consoles inside an office or room.	Used to connect your home or office to your Internet Service Provider (ISP).

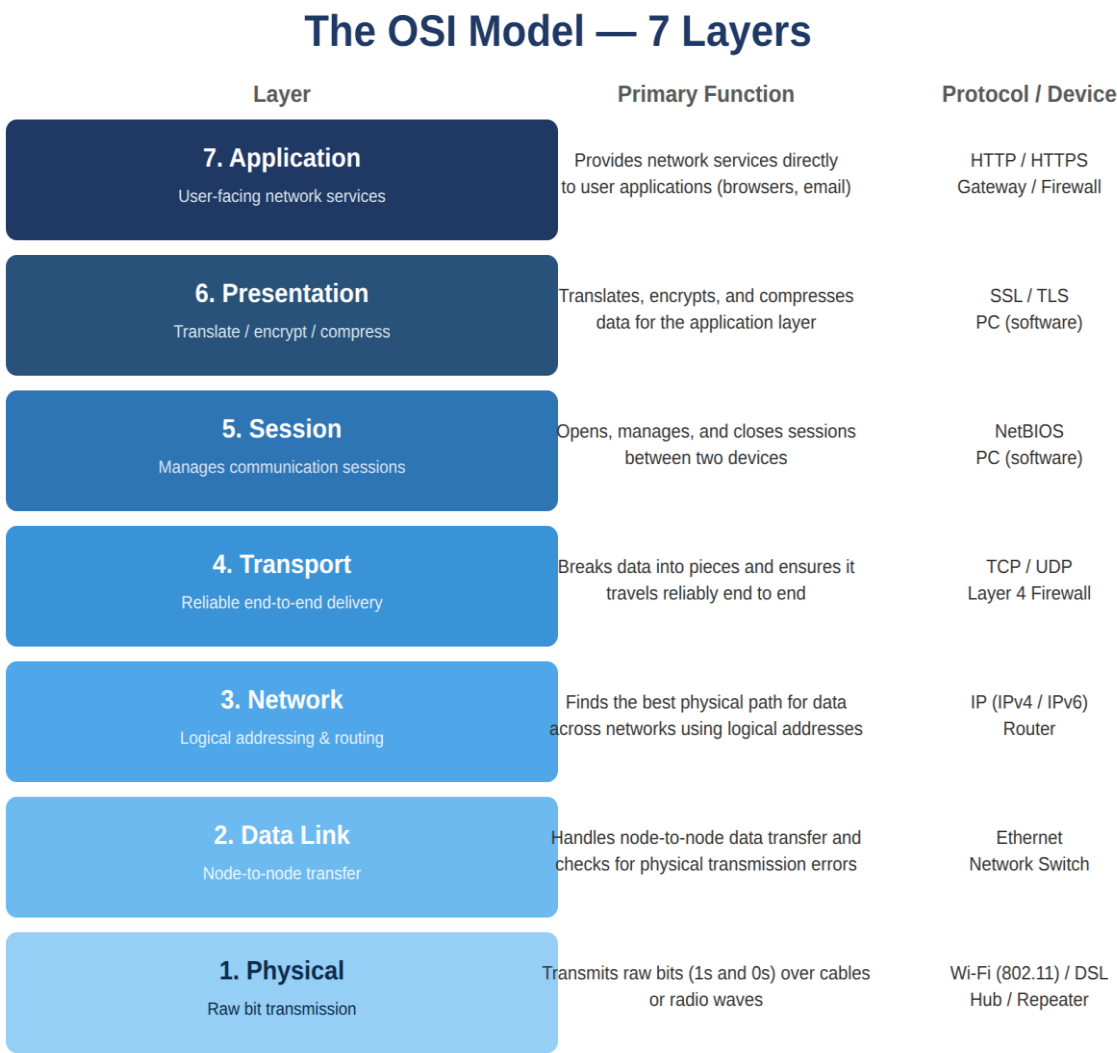
Q4. IPv4 vs IPv6

A comparison of IPv4 and IPv6 across six key differences, including one valid example address for each.

Feature	IPv4	IPv6
Address Size	32-bit address.	128-bit address.
Format	Written in numbers separated by dots (Decimal).	Written in numbers and letters separated by colons (Hexadecimal).
Number of Addresses	About 4.3 billion (Running out globally).	Practically infinite.
Security	Security features (IPsec) are optional.	Built-in native security features (IPsec is mandatory).
Configuration	Usually requires manual setup or a DHCP server.	Supports auto-configuration (devices configure themselves).
Valid Example	192.168.1.1	2001:0db8:85a3:0000:0000:8a2e:0370:7334

Q5. The OSI Model

All seven layers of the OSI Model, listed in the correct order. For each layer: its primary function, one commonly used protocol, and one networking device associated with that layer.



Model, with function, protocol, and device for each

1. Application Layer

- **Primary Function:** Provides network services directly to user applications (like web browsers or email clients).
- **Protocol:** HTTP / HTTPS
- **Device:** Gateway / Firewall

2. Presentation Layer

- **Primary Function:** Translates, encrypts, and compresses data so the application layer can understand it.
- **Protocol:** SSL / TLS
- **Device:** Gateway / PC (handled in software)

3. Session Layer

- **Primary Function:** Opens, manages, and closes communication sessions between two devices.
- **Protocol:** NetBIOS
- **Device:** Gateway / PC (handled in software)

4. Transport Layer

- **Primary Function:** Breaks data into pieces and ensures it travels reliably and accurately from end to end.
- **Protocol:** TCP / UDP
- **Device:** Layer 4 Firewall

5. Network Layer

- **Primary Function:** Finds the best physical path for data to travel across different networks using logical addresses.
- **Protocol:** IP (IPv4 / IPv6)
- **Device:** Router

6. Data Link Layer

- **Primary Function:** Handles node-to-node data transfer on the same network and checks for physical transmission errors.
- **Protocol:** Ethernet
- **Device:** Network Switch

7. Physical Layer

- **Primary Function:** Transmits raw bits (1s and 0s) over physical mediums like cables or radio waves.
- **Protocol:** Wi-Fi (802.11) / DSL
- **Device:** Hub / Network Cables / Repeater

Network Diagram

The diagram below illustrates how the concepts from this lab fit together in a typical home or office network: end devices connect to a Switch (Layer 2, using MAC addresses), which connects to a Router (Layer 3, using IP addresses) that provides the Default Gateway out to the Internet.

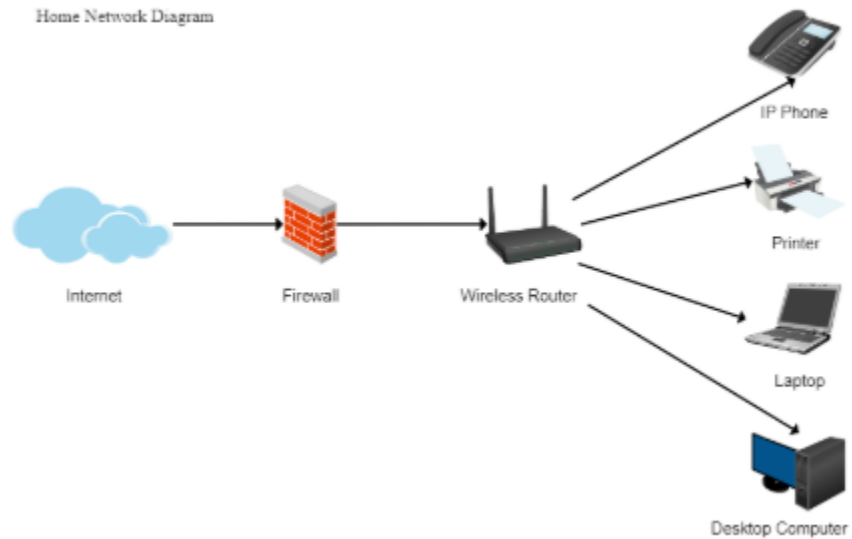


Figure 4 — Simplified network topology showing devices, Switch, Router, and Internet connection

Conclusion

This lab covered the core building blocks of computer networking, from identifying a machine's own network identity to understanding how data actually moves across the internet.

Commands such as `ipconfig /all`, `ping`, `tracert`, `nslookup`, and `arp -a` proved to be practical diagnostic tools: they reveal a device's local identity (IPv4 and MAC address), confirm connectivity and latency, trace the path packets take across multiple hops, resolve human-readable domain names into IP addresses, and map IP addresses to physical hardware on the local network.

Comparing routers and switches clarified how devices at different OSI layers serve different purposes — switches move frames between devices on the same local network using MAC addresses, while routers move packets between different networks using IP addresses. Similarly, contrasting IPv4 and IPv6 highlighted why the transition to IPv6 is necessary: IPv4's 32-bit address space is nearly exhausted, while IPv6's 128-bit space, built-in security, and auto-configuration support the continued growth of connected devices.

Finally, working through the OSI Model tied these concepts together into a single framework. Each of the seven layers — Physical, Data Link, Network, Transport, Session, Presentation, and Application — has a distinct responsibility, protocol, and associated device, and together they describe the complete journey data takes from one application to another across a network.

Overall, this exercise reinforced both the theoretical model (OSI) and the hands-on commands used daily to diagnose and understand real-world networks.